Literature Status: Nyström-SGM Beyond the Saturation Threshold

Run `fa_banach_001`, lane 2

11 August 2026

Status. `literature_already_answered`. This is an exact later-literature match, not a new result of the run.

Source question

Junhong Lin and Lorenzo Rosasco, *Optimal Rates for Learning with Nyström Stochastic Gradient Methods*, arXiv:1710.07797v1 (2017), Section 4, PDF p. 10, state that their NySGM bounds saturate at the source exponent $r = 1/2$ and conjecture:

> "by considering a different subsampling technique, we may get optimal error bounds even for $r \geq 1/2$."

Their optimal prediction-error rate, in the notation used there, is

$$n^{-\frac{2r+1}{2r+\gamma+1}}$$

up to logarithmic factors.

Supporting answer

Junhong Lin and Volkan Cevher, *Convergences of Regularized Algorithms and Stochastic Gradient Methods with Random Projections*, JMLR **21**(20):1–44 (2020), explicitly cites the source as "Lin and Rosasco (2017a)." Section 4, PDF pp. 12–14, gives the answer:

- Theorem 2 proves optimal projected-SGM rates with no upper restriction on its smoothness exponent s.
- Corollary 5 specifies the alternative approximate-leverage-score (ALS) Nyström subsampling scheme.
- Corollary 7 applies Theorem 2 when the projection and subsample size are those of Corollary 5.
- Remark 5(2) identifies the exact predecessor and says that the plain Nyström-SGM result of Lin–Rosasco (2017a) covered only $s \in [1/2, 1]$.

The notation aligns exactly. The 2017 condition $F_H = T^r h$ implies, for the prediction function $f_H = S_\rho F_H$,

$$f_H = L^{r+1/2} g,$$

so the 2020 exponent is $s = r + 1/2$. Setting $a = 0$ in Theorem 2 gives

$$n^{-\frac{2s}{2s+\gamma}} = n^{-\frac{2r+1}{2r+\gamma+1}},$$

the same optimal rate. The formerly saturated range $r > 1/2$ is precisely $s > 1$, which Theorem 2 and Corollary 7 include. Thus the later authors knew the predecessor and supplied the requested different Nyström subsampling route with optimal rates.

Scope and search evidence

The answer is in the source problem's least-squares Hilbert/RKHS setting and under the assumptions stated in the 2020 Theorem 2 and ALS Corollary 5. It does not resolve the source paper's separate broad future directions on joint cross-validation of every algorithmic parameter, unbounded kernels, preconditioning, random features, or acceleration.

The bounded search checked the run indexes, the exact arXiv id and title, "NySGM saturation zeta subsampling," "Nyström stochastic gradient optimal rates," and the citation trail from arXiv:1710.07797. The JMLR paper is decisive because it cites the source, describes its limited smoothness range, and extends that exact algorithmic line.

Local evidence

- `source_paper.pdf`: arXiv:1710.07797v1.
- `supporting_paper_jmlr_2020_19-083.pdf`: JMLR 21(20), 2020.
- Ledger: `ledger/results/1710.07797_nysgm_saturation_answered_2020.json`.

References

[1] J. Lin and L. Rosasco, *Optimal Rates for Learning with Nyström Stochastic Gradient Methods*, arXiv:1710.07797v1, 2017.

[2] J. Lin and V. Cevher, *Convergences of Regularized Algorithms and Stochastic Gradient Methods with Random Projections*, Journal of Machine Learning Research 21(20) (2020), 1–44.